

# Start here: choose your project

Begin with the Edge Starter. Learn what one change does, test it, then add your own strategy. All links open editable MakeCode projects with their extensions included.

## 1 / Edge Starter v2.0 - competition starting point

Waits for START, drives slowly and recovers from black. Set your team number. The referee assigns the match lights.

<https://makecode.microbit.org/S04419-26199-02954-87785>



## 2 / Color Lab v2.0 - optional lesson

Motors stay stopped. Compare a brightness threshold with measured red, green and blue values. Instructions: page 5.

<https://makecode.microbit.org/S51417-07093-46392-81607>



## 3 / Ultrasonic Lab v2.0 - optional lesson

Measure distance first; try a wheels-raised response later. Uses A/B practice controls, not referee radio. Instructions: page 6.

<https://makecode.microbit.org/S29595-96351-49856-35763>



## 4 / Blank Tournament Template v2.0 - write a strategy

Wukong version. Tournament controls are ready; YOUR\_robotCode is empty. No edge detection is included. Instructions: page 7.

<https://makecode.microbit.org/S37311-09323-32132-28628>



**Open:** click a link or scan its QR code, choose **Edit Code**, then **Blocks**. Rename your copy, for example **Team 123 - Summit**. Use internet for the first load; no Chrome extension is needed.

**Route:** setup and controls (p. 2) → stationary edge checks (p. 3) → movement and tuning (p. 4). Labs are optional and replace the program on your micro:bit: restore your competition project afterward.

Different chassis or motor board? Use the separate **Custom Robot Guide**; kit pin assignments and motor settings are not universal.

# Set up, download, start and stop

1. In Edge Starter, set **myTeam** in **on start** to your assigned number. Leave the three function calls underneath it in place.
2. With power off, check plugs, motor connections and sensor placement using the Build & Wiring Guide. Support the robot with both wheels clear. Keep fingers and loose wires away from moving parts.
3. Connect a USB **data** cable. Click **Download**. Pair the micro:bit when prompted, or use **Download as File** and copy the .hex file to the **MICROBIT** drive. Wait for completion.
4. Power the Wukong board. The robot should remain still with a small diamond on its display. Complete page 3 before starting movement.

| Control               | Expected result                                                                                                                   |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Robot button A        | Five-second practice countdown, then movement.                                                                                    |
| Robot button B        | Stop, cancel any countdown, and stay stopped until reset.                                                                         |
| Referee buttons A+B   | Assign red/blue lights without starting.                                                                                          |
| Referee buttons A / B | A counts down and starts both; B stops both.                                                                                      |
| Next match            | After referee STOP, wait for its X to become a small diamond, then reset BOTH robots. Resetting earlier can receive another STOP. |

## Know which blocks are yours

| Area                                                         | What to do                                                                                                 |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| EDIT_settings                                                | Keep the defaults first. Later, change one setting and retest.                                             |
| EDIT_robotBehavior                                           | Optional strategy belongs in the LAST else, below the edge checks.                                         |
| TOURNAMENT and HELPER functions; button/radio/forever blocks | Keep unchanged. Folded means hidden detail, not disabled code. The TOURNAMENT label is not an editor lock. |

A function is a named group of blocks; a **call** block runs it. You can use prepared functions without understanding every block inside them. If controls fail, switch off drive power.

# Test the edge decision before driving

**Use Edge Starter, not Color Lab.** Keep the wheels raised and do not press A. These readings continue while the robot waits for START.

1. In Chrome or Edge, use the Download menu to **Connect Device**. Open **Show data** → **Device**, not Simulator. Look for light, red, green, blue and edge\_detected.
2. Hold the mounted sensor over black. Record ten or more readings at normal height, then at the greatest realistic tilt. Expect **edge\_detected = 1**.
3. Repeat over dark blue, especially the raised angle after the bump. Expect **edge\_detected = 0**. Then check red, yellow, green and white: all should read 0.
4. Repeat with both red and blue match lights, assigned by the referee. Keep mounting height, sensor settings and lighting consistent.
5. If a test fails, change a threshold only from your measured ranges, download again, and repeat every color. A zero light value is unusable: check the sensor and wiring first.

| Reading                                            | Edge Starter decision                                                      |
|----------------------------------------------------|----------------------------------------------------------------------------|
| light = 0                                          | Stop movement. No usable floor reading.                                    |
| Blue exceeds BOTH red and green by blueMargin      | Do not begin a new black-edge response. An active recovery still finishes. |
| light is below blackThreshold, and not strong blue | Black edge: reverse, then turn when the floor clears.                      |

## Why brightness alone is not enough

On the tested robot, raised blue reached light 36-37 and black reached 27-43. The ranges overlap. The starter combines brightness with a blue-channel comparison. Defaults **64** and **15** are starting values, not calibration for every robot.

## Record your acceptance test

Black: normal light \_\_\_\_\_ / tilted light \_\_\_\_\_ / flag \_\_\_\_\_  
 Blue: normal light \_\_\_\_\_ / tilted light \_\_\_\_\_ / flag \_\_\_\_\_  
 All other rings checked? \_\_\_\_\_ Under both match lights? \_\_\_\_\_

# Test movement, then tune one setting

Only proceed after the stationary edge checks pass. Keep the supplied low speeds for the first trial.

1. With wheels raised and a non-black sample under the sensor, press A. After five seconds, check that both wheels would drive forward. Press B.
2. Reset and start again. Move a black sample under the sensor. Expect reverse for at least 600 ms. If black remains after one second, the wheels hold; after it clears, they turn and resume.
3. Test B during the countdown, forward motion, reversing and turning. It must stop and remain stopped. Black detected during a turn should begin a new backup.
4. Test straight and diagonal approaches on the practice Hill, with a soft catch area below. Change one value, predict the effect, download and test again.

| EDIT_settings value | Changes this one thing                                                |
|---------------------|-----------------------------------------------------------------------|
| cruiseSpeed = 20    | Normal forward motor setting.                                         |
| reverseSpeed = 20   | Backup motor setting.                                                 |
| turnSpeed = 20      | Turning motor setting.                                                |
| reverseMs = 600     | Minimum backup duration, in milliseconds.                             |
| maxReverseMs = 1000 | Longest continuous backup while black remains; then hold until clear. |
| turnMs = 350        | Turn duration. This does not guarantee 180 degrees.                   |
| blackThreshold = 64 | Brightness below which black is possible. Tune from page 3 readings.  |
| blueMargin = 15     | How far blue must exceed both other channels for the blue exception.  |

**My prediction:** \_\_\_\_\_

**One setting changed:** \_\_\_\_\_

**Result and next step:** \_\_\_\_\_

## Limits to test on your robot

One front sensor cannot see the rear edge during a reverse. Momentum, slope and mounting height affect stopping distance. Edge avoidance reduces falls; it cannot guarantee recovery from every approach or push.

# Color Lab: read the floor

**Goal:** learn what the floor sensor actually measures. The motors stay stopped; no A press is needed. Open **Color Lab v2.0** from page 1.

| Change in EDIT_settings  | What you see in Show data → Device                                                                                           |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------|
| testMode = BLACK         | light and black_detected. A valid light reading below blackThreshold gives 1; otherwise 0. This is a brightness-only lesson. |
| testMode = COLORS        | light, red, green and blue. Measurements do not automatically label a ring.                                                  |
| matchColor = RED or BLUE | The kit lights change for this test only; this is not a tournament color assignment.                                         |

1. Keep **BLACK** for the first lesson. Download, pair USB, and open Device data. Compare black and raised blue. If both trigger, explain why one brightness threshold cannot separate them.
2. Set **testMode = COLORS** using uppercase letters and download again. Record at least ten readings per ring at normal height and realistic tilt. Use ranges, not a single reading.
3. Repeat with the other matchColor light setting. Look for a channel comparison that stays true for one ring and false for all others.
4. Try your proposed rule with a serial label or display icon before adding movement to your competition program.

**sensor\_valid = 0:** ignore current/old color values and check the sensor, wiring and lighting. Channel values are ratios scaled by 255; they are not guaranteed to be limited to 0-255.

| Ring / angle | Light range | Red range | Green range | Blue range |
|--------------|-------------|-----------|-------------|------------|
| _____        | _____       | _____     | _____       | _____      |
| _____        | _____       | _____     | _____       | _____      |

Repeat this record for black, blue, red, yellow, green and white, including tilt.

**Rule to test:** \_\_\_\_\_

**Which other ring could fool it?** \_\_\_\_\_

For example, "green exceeds red and blue" is a hypothesis, not a calibrated green detector. Verify your rule on every ring. Use Edge Starter for the actual black-edge acceptance test on page 3.

# Ultrasonic Lab: detect a nearby target

**Goal:** measure distance, then test one simple response. Open **Ultrasonic Lab v2.0**. Its code uses TRIG = P1, ECHO = P2 and centimeters. Wukong signal pins are labeled **1** and **2**.

## First: SENSOR mode

1. Keep **testMode = SENSOR** in EDIT\_settings and download. The motors stay stopped, including if A is pressed. No A press is needed for readings.
2. Pair USB and open **Show data** → **Device**. Look for **distance\_cm**. Center a flat card in front of the sensor at about 10, 20 and 30 cm, measured from its front face.
3. Check that the reading changes sensibly with distance. Then try an angled card. Zero means no valid echo in this lesson; it is never treated as a close target.

| Measured target distance | Reported distance_cm |
|--------------------------|----------------------|
| 10 cm                    | _____                |
| 20 cm                    | _____                |
| 30 cm                    | _____                |

## Next: MOVE mode, wheels raised

1. Set **testMode = MOVE**, download and keep the robot supported. It must remain still before starting.
2. Press A once and wait five seconds. With **targetDistance = 25**, a valid reading greater than 0 and less than 25 commands forward movement at **approachSpeed = 20**. At 25 or farther, or at zero, it stops on the next reading.
3. Press B, then bring the card close again. The wheels must stay stopped. Reset before another trial. There is also a three-minute cutoff.
4. Change only targetDistance to 15. Predict which of your three card positions will trigger movement, then download and retest.

### Practice lesson only

No edge detection or referee radio is included. Do not run MOVE mode on the raised Hill. Restore your competition program after the lab. If readings are always zero, check wiring, power compatibility and target position before changing thresholds.

# Write your strategy and check it

Choose either route. Both use the same tournament controls; you do not need to rewrite the start/stop system.

**Extend Edge Starter:** use the LAST else in EDIT\_robotBehavior. Keep the floor checks and recovery above it.

**Blank Wukong template:** set myTeam, then write inside YOUR\_robotCode. Add sensor setup after TOURNAMENT\_setup if needed. You must supply edge detection.

## A simple decision order

**Read → decide → drive → repeat**

Invalid floor reading? Stop. Edge or recovery active? Recover.

Otherwise: valid nearby target? Approach. No target? Search.

Use **Advanced → Functions → call TOURNAMENT\_drive** for every movement. On this kit, its inputs **20, -20** drive forward and **0, 0** stop. Verify direction with wheels raised. Keep the function definition unchanged.

Keep each pass short: no extra forever loop inside your behavior and no long pauses that delay checks. Use elapsed-time variables for multi-step moves. Robots receive official radio commands; they do not transmit.

## Pass these checks before a match

| Check                                 | Pass condition                                                                                      |
|---------------------------------------|-----------------------------------------------------------------------------------------------------|
| Power on / reset                      | No motion before START or completed A countdown.                                                    |
| A then B during countdown             | No later movement; reset required.                                                                  |
| Referee test                          | Correct team ID, correct red/blue assignment, both robots start and stop. Repeat at field distance. |
| B or referee STOP during any behavior | Stops immediately and stays stopped until reset.                                                    |
| Three-minute run                      | Local timer stops the robot without needing referee STOP.                                           |
| Target and edge together              | An opponent cannot override edge recovery.                                                          |

**Only one robot starts?** Referee stops both. Wait for the referee diamond, reset both and replay. Do not start the late robot mid-match.

[MakeCode USB/data help](#) | [Current game manual](#) | [Color extension reference](#)